Guideline for Estimating Design Hours - Bareshaft Pumps

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1. Purpose & Scope

This procedure provides a standard guideline for the correct and consistent compilation of budgeted engineering hours related to the design and development of bareshaft centrifugal pumps. It is meant to support the use of the official Excel tool **Mod.555**.

Thia procedure applies to all personnel involved in planning and recording budget hours for new projects, product customizations, or development activities on OH (Overhung), BB (Between Bearings) and VS (Vertical) bareshaft centrifugal pumps.

This procedure is applicable to Nova Milanese and La Spezia plants.

2. Definitions

BB: Between Bearings pumps

BoM: Bill of Materials

CAD: Computer-Aided Design

FEM: Finite Element Method

OH: Overhung pumps

PE: Product Engineering

R&D: Research and Development

3. General Guidelines

The Excel template includes a structured breakdown of engineering activities, subdivided into product types, technical functions, and project phases. Each row represents a task or a set of activities to be estimated.

The goal is to ensure uniformity and traceability in budget estimations to support project planning, cost monitoring, and resource allocation.

4. Budget Structure and Sections

The workbook is composed of the following major areas:

- “Data Input” Form– The data entering form (activated by pressing the “ACTIVATE INPUT FORM” button).

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- "MASTER" Sheet – Performs the calculation of the design hours.
- "REPORT" Sheet – Summary of all the input parameters and of the calculated budget hours divided by design steps and by network phases.
- "GANTT" and "GANTT_D" Sheets – Provide preliminary GANTT diagrams for all the engineering activities, on weekly or daily basis.
- "Static analysis" Sheet – Performs the calculation to evaluate the recommended complexity level of the static analysis.
- "Dynamic analysis" Sheet – Performs the calculation to evaluate the recommended complexity level of the dynamic analyses.
- Sheets with Design hours – Sheets with tables of the estimated hours for all the design tasks.
- "NETWORK_Structure" Sheet – Contains detailed information on the SAP network structure for design activities.

5. Input Instructions

1. Open **Mod.555** and enable macros.
2. Go to "MASTER" sheet and push "ACTIVATE INPUT FORM".
3. Input the data into the four pages of the form as described below.

The first page is labelled "**General**" (*Fig. 1*), and the following information are required:

- **Reference Project / Opportunity**: Indicate the project WBS reference if it is in the execution phase or the opportunity number if it is in the quoting phase.
- **Pump Configuration**: Specify if the centrifugal pump is vertical or horizontal.
- **Pump Family (API)**: Select the pump family from the drop-down list according to API 610 standard.
- **Pump Type**: Specify the name of the pump according to internal nomenclature.
- **Energy Level / Design Complexity**: Specify the energy level or design complexity using the drop-down menu and choose from the three possibilities (Low; Medium; High). The Energy Level must be categorized according to **SOP-489** and by means of **Mod.469**.
- **Pump Origin**: Specify the pump origin selecting from the drop-down list, one of the three possibilities:

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- Existing: the design of all components already exists.
- Scaled: the design of one or more components must be scaled from an existing design.
- New: the design of one or more components must be considered as new.

- **Reset**: Use this button to clear all the cells.

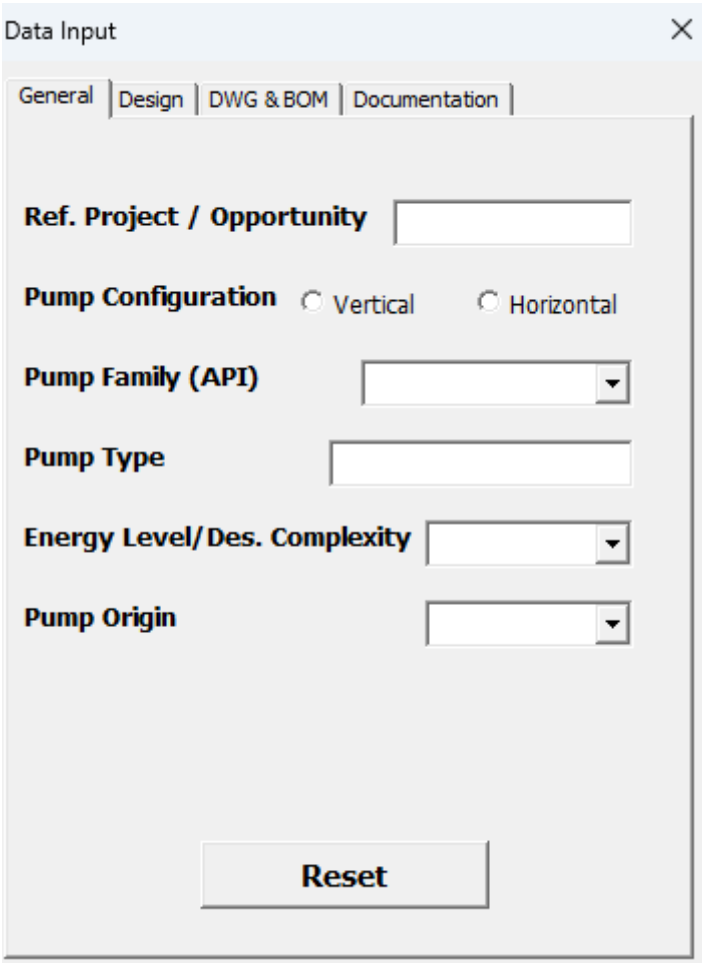

Fig. 1: "General" page.

The second page is labelled "**Design**" (*Fig. 2*), and the following information are required:

- **Pump Hydraulic Design**: Indicate which kind of hydraulic design is required among the four possibilities:
  - NO: no hydraulic design is required (to be selected if pump origin is existing or scaled).
  - Single Part: hydraulic design is needed only for one component (impeller, diffuser, etc.).

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- Single Stage: hydraulic design is needed for all the components of one stage.
- Full: hydraulic design is needed for all hydraulic components.

- **Intake/ Piping Redesign**: specify if hydraulic design is requested for any part of the pumping system other than the pump, for example the intake basin or suction tank of vertical pumps or the suction piping. The design of the intake basin or tank is typically requested by the client to prevent swirl/vortex problems at the pump's suction bell of vertical pumps. Piping design analysis can be requested by the client or can be necessary if nonstandard configurations are applied on site at the pump's suction line.

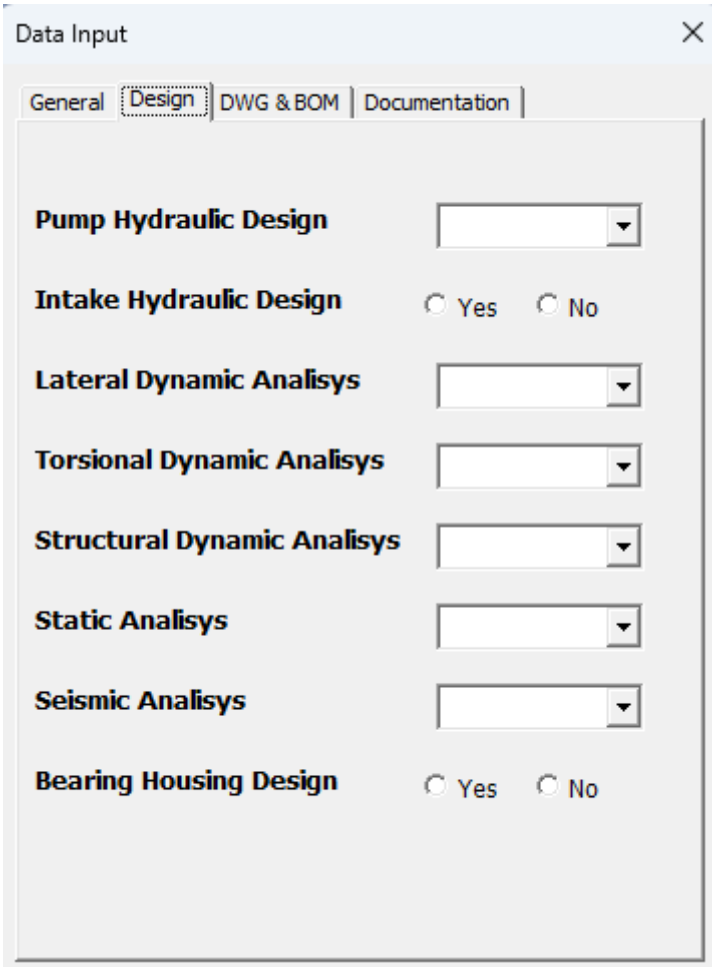

Fig. 2: "Design" page.

- **Lateral Dynamic Analysis, Torsional Dynamic Analysis and Structural Dynamic Analysis**: Indicate if a lateral, torsional or structural dynamic analysis is

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needed and which complexity level is required, using the drop-down list with four possibilities (NO, Level 1, Level 2, Level 3).

The recommended complexity level of the dynamic analyses must be evaluated with the "dynamic analysis" sheet (*Fig. 3*), by entering the required data in the yellow fields.

	dynamic analysis				
			Lateral	Torsional	Structural
1	pump type				
2	n_{max} [rpm]		0	0	0
3	S.G. [-]		2	0	0
4	Driver support		0	0	0
5	Flexible foundation		0	0	0
6	Driver		0	0	0
7	Train with 3 or more elements		0	0	0
8	Variable speed		0	0	0
9	Power [kW]		1	1	1
10	$n_{blade} \leq 3$		0	0	0
		U [-]	**4**	**2**	**1**
	existant design	**R [-]**	**0**	**0**	**0**
		RUN [-]	**0**	**0**	**0**
	Level		**Level 0**	**Level 0**	**Level 0**

Fig. 3: "Dynamic Analysis" Sheet.

A more detailed assessment can be made through **Mod.401**, according to the operating procedure **G10.00.09.** This method requires more technical information which may not be available at an early stage or without having specific technical competences.

- **Static Analysis**: Indicate if a static analysis is needed and which complexity level is required, using the drop-down list with four possibilities (NO, Level 1, Level 2, Level 3).

  The recommended complexity level of the static analyses must be evaluated with the "static analysis" sheet (*Fig. 4*), by entering the required data in the yellow fields.

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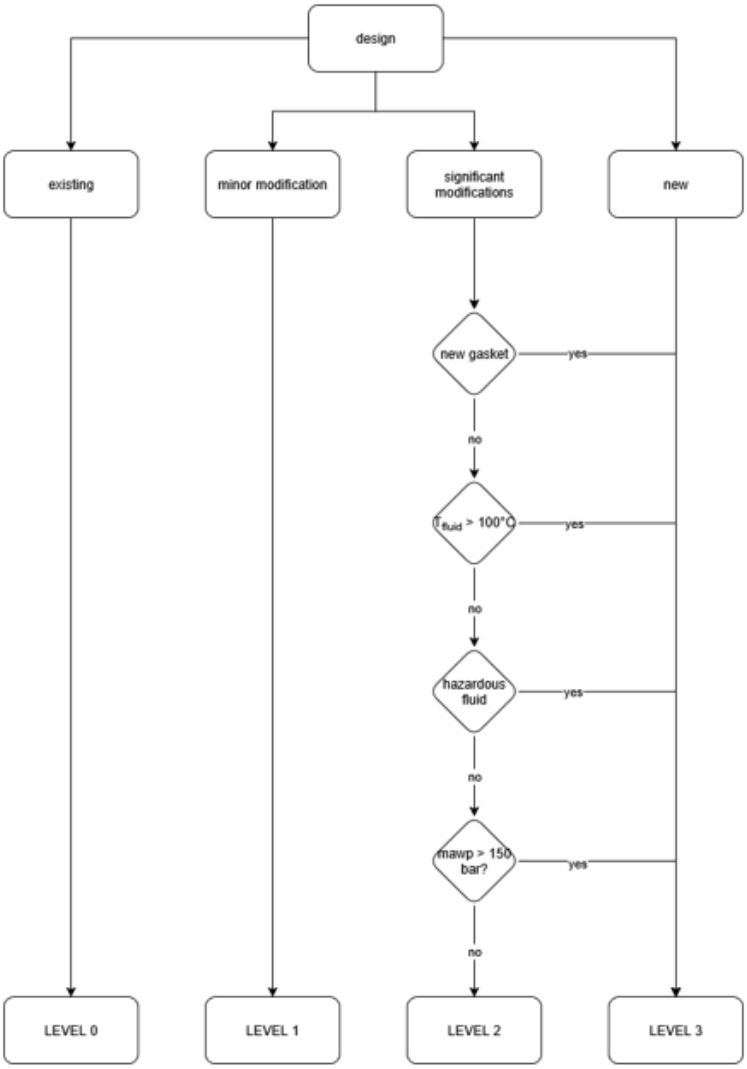
A more detailed assessment can be made according to the operating procedure **G10.00.21**. This method requires more technical information which may not be available at an early stage or without having specific technical competences.



design	significant modification
new gasket	no
T_{fluid} > 100°C	no
hazardous fluid	no
mawp > 150 bar	no
Level	Level2

Fig. 4: “Static Analysis” Sheet.

- **Seismic Analysis**: Indicate if a seismic analysis is needed and which complexity level is required, using the drop-down list with four possibilities (NO, Level 1, Level 2, Level 3).

  The seismic analysis is usually performed only if requested by the customer when the pump unit is installed in a seismic area.

  The complexity level is usually specified in the client specifications, otherwise the following criteria may be applied:

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- Level 1: static equivalent analysis, recommended for horizontal pumps.
- Level 2: response spectra analysis, recommended for vertical pumps.
- Level 3: transient analysis (rarely applicable).

- **Bearing Housing Redesign**: Indicate whether an existing bearing housing will be used (no additional hours required) or if a new design is necessary for the scope.

The third page is labelled "**DWG & BOM**" (*Fig. 5*), and the following information are required:

- **Number of Impeller Design**: Indicate the number of different hydraulics included in the pump, for which different drawings must be issued.
- **Integral Bosses Modification:** Indicate whether the integral bosses modification is required. The integral bosses are required for all horizontal pumps. The modification is needed for all the existing designs where integral bosses are not present.

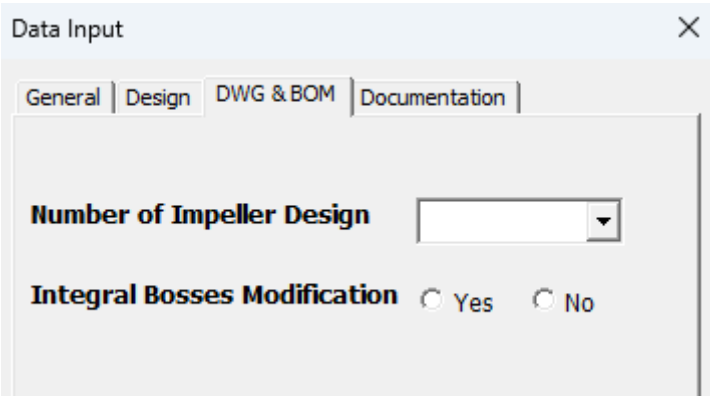

Fig. 5: "DWG & BOM" page.

The fourth page is labelled "**Documentation**" (*Fig. 6*), and the following information are required:

- **Project Calculation Reports:** Indicate the required set of client documentation to be provided by Product Engineering, choosing from the drop-down list one of three possibilities:
  - NO: no documentation is required.
  - Standard: only lateral analysis report and bearing data sheet are required.

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- Full: complete set of calculation reports: lateral analysis, torsional analysis, pump casing structural FEM analysis, vertical pump modal FEM analysis, bearing data sheet.

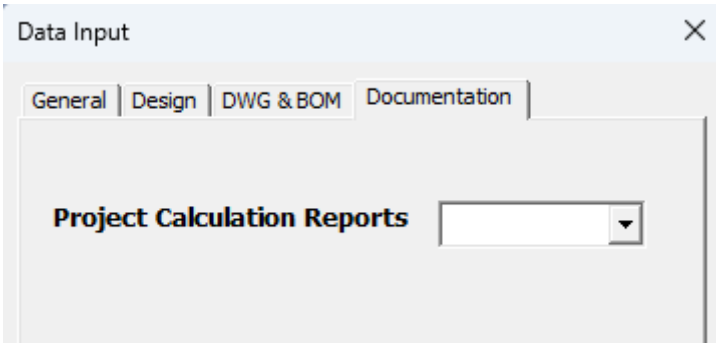

Fig. 6: “Documentation” page.

If special design features, not included in the input template, are applicable to the project (i.e. heating chamber, coke crusher, etc.), contact Product Design for estimation of the additional hours needed.

6. Export of data to Technical Review

After completing all input data, save a copy of the file using the filename “Design_Budget_Hours_[WBS].xlsxm”), in the “18_Technical_Review” sub-folder located in the T:\IT005\JM\[year]\[project] directory.

After saving the file, click on the “Export to Technical Summary“ button in the “REPORT” sheet to export the budget hours report and the preliminary GANTT to the job technical summary file.

7. References

- **SOP-489** - High-Medium-Low Energy pumps categorization
- **Mod.469** - High-Medium-Low Energy pumps categorization
- **G.10.00.09** - Guidelines for dynamic analysis complexity level definition
- **Mod.401** - ANSI HI 9.6.8 - Dynamic analysis complexity level definition
- **G.10.00.21** - Guidelines for static analysis complexity level definition

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8. Annex A – Budget Entry Example

Example of application to WBS 087076.

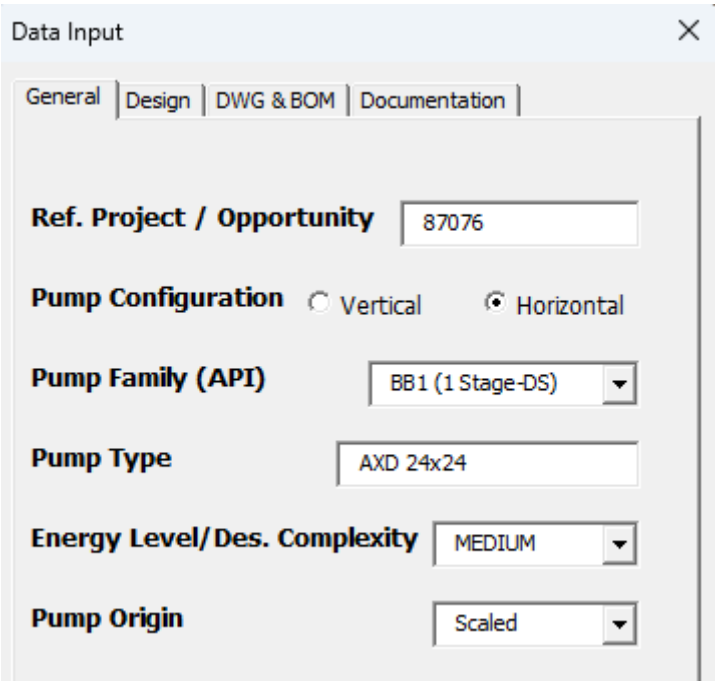

Fig. 7: Example of data input – "General" page.

	dynamic analysis				
			Lateral	Torsional	Structural
1	pump type	BB			
2	n_{max} [rpm]	990	0	0	0
3	S.G. [-]	0,991	0	0	0
4	Driver support	pump	0	0	0
5	Flexible foundation	yes	0	0	1
6	Driver	asynchronous electric motor	0	0	0
7	Train with 3 or more elements	no	0	0	0
8	Variable speed	no	0	0	0
9	Power [kW]	315	1	1	1
10	$n_{blade} \leq 3$	no	0	0	0
		U [-]	**2**	**2**	**2**
	significant modification	**R [-]**	**10**	**10**	**10**
		RUN [-]	**20**	**20**	**20**
	Level		**Level 1**	**Level 1**	**Level 1**

Fig. 8: Example of complexity level assessment for dynamic analyses.

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design	significant modification
new gasket	yes
T_{fluid} > 100°C	no
hazardous fluid	no
mawp > 150 bar	no
Level	*Level 3*

Fig. 9: Example of complexity level assessment for static analysis.

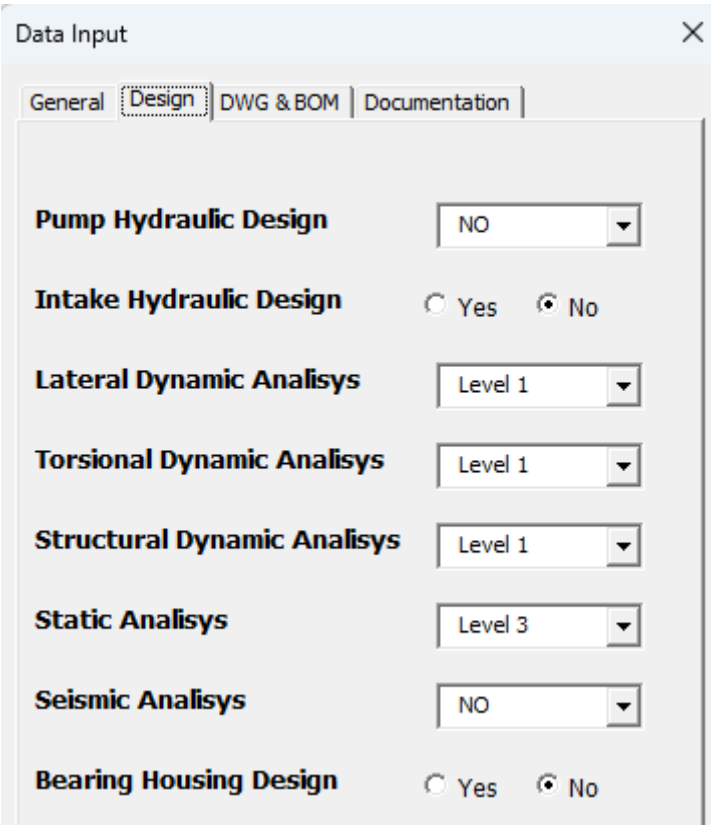

Fig. 10: Example of data input – “Design” page.

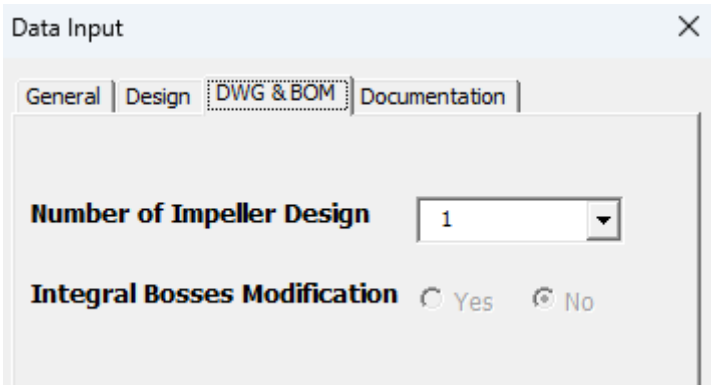

Fig. 11: Example of data input – “DWG & BOM” page.

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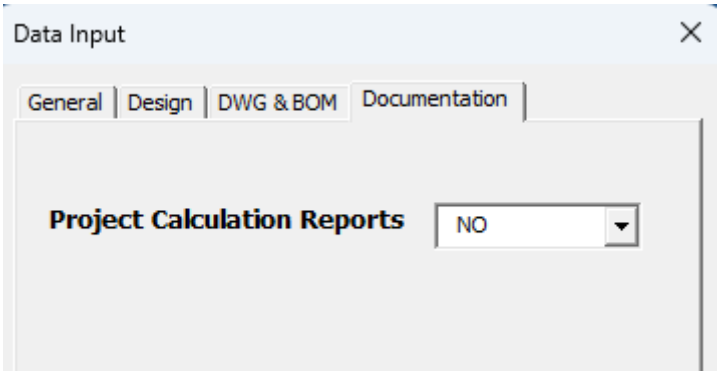

Fig. 12: Example of data input – "Documentation" page.

INPUT DATA	
SECTION #1: GENERAL	
Pump Configuration	Horizontal
Pump Family	BB1 (1 Stage-DS)
Pump Type	AXD 24x24
Reference Project/Opportunity	087076
Energy Level / Design Complexity	MEDIUM
Performance Curve Origin	Scaled
SECTION #2: DESIGN	
Hydraulic Design: Pump	NO
Hydraulic Design: Intake/ Piping	NO
Dynamic Analysis: Lateral	Level 1
Dynamic Analysis: Torsional	Level 1
Dynamic Analysis: Structural	Level 1
Static Analysis	Level 3
Seismic Analysis	NO
Bearing Housing Design	NO
SECTION #3: DRAWINGS AND BOM	
Number of impeller design	1
Integral bosses modification	NO
SECTION #4: DOCUMENTATION	
Project calculation reports	NO

ENGINNERING HOURS ESTIMATION			
Task	**Description**	**Hours**	**Days**
DESIGN		**152,5**	**21,8**
R&D		*72,5*	*10,4*
Product Design		*42*	*6,0*
Product Technology		*38*	*5,4*
DWGS AND BOM		**312**	**44,6**
D001	*PATTERN 3D CASING DWG*	*136*	*19,4*
D002	*PATTERN 3D COVER DWG*	*0*	*0,0*
D003	*PATTERN 3D IMPELLER DWG*	*32*	*4,6*
D004	*PATTERN 3D DIFFUSER DWG*	*0*	*0,0*
D005	*PATTERN 3D OTHER DWG **	*0*	*0,0*
D010	*CASTINGS RELEASE PREX E-BOM*	*20*	*2,9*
D015	*PREX E-BOM COMPLETION**	*44*	*6,3*
D018	*PREX VERT & WELD STRUCTURE E-BOM*	*0*	*0,0*
D020	*BSHP E-BOM*	*80*	*11,4*
R&D		*0*	*0,0*
Product Design		*312*	*44,6*
Product Technology		*0*	*0,0*
DOCS		**28**	**4,0**
R&D		*0*	*0,0*
Product Design		*28*	*4,0*
Product Technology		*0*	*0,0*
Tot. Engineering Hours		**492,5**	**70,4**
Total R&D		*72,5*	*10,4*
	Fuid Dynamics Design	0	0,0
	Advanced Mechanical Design	72,5	10,4
Total Product Design		*382*	*54,6*
Total Product Technology		*38*	*5,4*

Fig. 13: Example of engineering hours estimation report.

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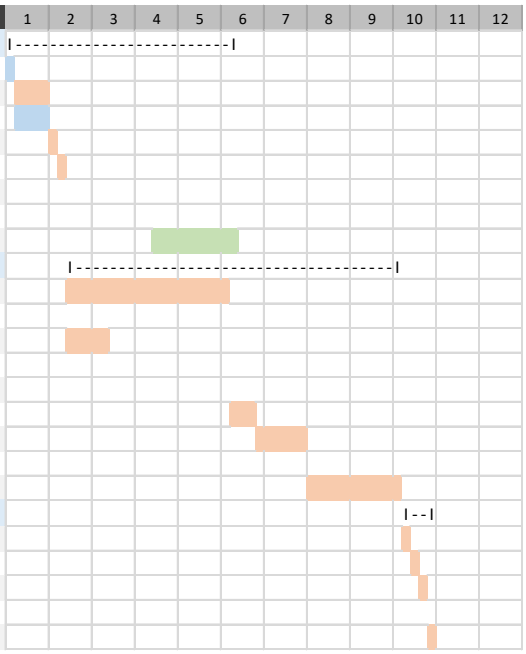
Reference Project/Opportunity: **087076**
Pump Type: **AXD 24x24**
Pump Family: **BB1 (1 Stage-DS)**

Product Technology
Product Design
Fuid Dynamics Design
Advanced Mechanical Design

Task	Owner	Days	Weeks
DESIGN			
Technical Review	Product Technology	1	1
Study Drawing	Product Design	4	1
Preliminary Sizing	Product Technology	4	1
Seal Chamber Drawing	Product Design	1	1
dummy GAD	Product Design	1	1
Fluid Dynamics Analysis	Fuid Dynamics Design	0	
Special Design: Intake/ Piping	Fuid Dynamics Design	0	
Structural Analysis	Advanced Mechanical Design	10	3
DWGS AND BOM			
D001 - PATTERN 3D CASING DWG	Product Design	19	4
D002 - PATTERN 3D COVER DWG	Product Design	0	
D003 - PATTERN 3D IMPELLER DWG	Product Design	5	1
D004 - PATTERN 3D DIFFUSER DWG	Product Design	0	
D005 - PATTERN 3D OTHER DWG *	Product Design	0	
D010 - CASTINGS RELEASE PREX E-BOM	Product Design	3	1
D015 - PREX E-BOM COMPLETION*	Product Design	6	2
D018 - PREX VERT & WELD STRUCTURE E-BOM	Product Design	0	
D020 - BSHP E-BOM	Product Design	11	3
DOCS			
Sectional Drawing and Part List	Product Design	1	1
Tightening Sequence	Product Design	1	1
Assembly Drawing	Product Design	1	1
Project calculation reports	Product Technology	0	
Check List	Product Design	1	1



Total workload	**70**
Working days	**50**
Calendar days	**68**
Calendar weeks	**10**

Fig. 14: Example of GANTT diagram.

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